

Publications by Model: MOMO-CHEM

Total Publications: 45 (sorted newest to oldest)

1. The Capability of Deep Learning Model to Predict Ozone Across Continents in China, the United States and Europe

Authors: Weichao Han (University of Science and Technology of China); Tai-Long He (University of Washington, University of Toronto); Zhe Jiang (University of Science and Technology of China); Rui Zhu (University of Science and Technology of China); Dylan Jones (University of Toronto); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yanan Shen (University of Science and Technology of China)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 50, Issue 24

References: 40

Cited by: 3 publications

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Open Access: N/A

Abstract: Data-driven methods have been extensively applied to predict atmospheric compositions. Here, we explore the capability of a deep learning (DL) model to make ozone (O₃) predictions across continents in China, the United States (US) and Europe. The DL model was trained and validated with surface O₃ observations in China and the US in 2015-2018. The DL model was applied to predict hourly surface O₃ over three continents in 2015-2022. Compared to baseline simulations using GEOS-Chem (GC) model, our analysis exhibits mean biases of 2.6 and 4.8 mug/m³ with correlation coefficients of 0.94 and 0.93 (DL); and mean biases of 3.7 and 5.4 mug/m³ with correlation coefficients of 0.95 and 0.92 (GC) in Europe in 2015-2018 and 2019-2022, respectively. The comparable performances between DL and GC indicate the potential of DL to make reliable predictions over spatial and temporal domains where a wealth of local observations for training is not available.

2. A simplified non-linear chemistry transport model for analyzing NO₂ column observations: STILT-NO_x

Authors: Dien Wu; Joshua L. Laughner; Junjie Liu; Paul I. Palmer; John C. Lin; Paul O. Wennberg

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-6161-2023

URL: <https://doi.org/10.5194/gmd-16-6161-2023>

Publication Date: 2023-11-01

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References: 90

Cited by: 1 publications

Language: en

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Open Access: N/A

Abstract: . Satellites monitoring air pollutants (e.g., nitrogen oxides; NO_x = NO + NO₂) or greenhouse gases (GHGs) are widely utilized to understand the spatiotemporal variability in and evolution of emission characteristics, chemical transformations, and atmospheric transport over anthropogenic hotspots. Recently, the joint use of space-based long-lived GHGs (e.g., carbon dioxide; CO₂) and short-lived pollutants has made it possible to improve our understanding of emission characteristics. Some previous studies, however, lack consideration of the non-linear NO_x chemistry or complex atmospheric transport. Considering the increase in satellite data volume and the demand for emission monitoring at higher spatiotemporal scales, it is crucial to construct a local-scale emission optimization system that can handle both long-lived GHGs and short-lived pollutants in a coupled and effective manner. This need motivates us to develop a Lagrangian chemical transport model that accounts for NO_x chemistry and fine-scale atmospheric transport (STILT-NO_x) and to investigate how physical and chemical processes, anthropogenic emissions, and background may affect the interpretation of tropospheric NO₂ columns (tNO₂). Interpreting emission signals from tNO₂ commonly involves either an efficient statistical model or a sophisticated chemical transport model. To balance computational expenses and chemical complexity, we describe a simplified representation of the NO_x chemistry that bypasses an explicit solution of individual chemical reactions while preserving the essential non-linearity that links NO_x emissions to its concentrations. This NO_x chemical parameterization is then incorporated into an existing Lagrangian modeling framework that is widely applied in the GHG community. We further quantify uncertainties associated with the wind field and chemical parameterization and evaluate modeled columns against retrieved columns from the Tropospheric Monitoring Instrument (TROPOMI v2.1). Specifically, simulations with alternative model configurations of emissions, meteorology, chemistry, and inter-parcel mixing are carried out over three United States (US) power plants and two urban areas across seasons. Using the U.S. Environmental Protection Agency (EPA)-reported emissions for power plants with non-linear NO_x chemistry improves the model-data alignment in tNO₂ (a high bias of $\leq 10\%$ on an annual basis), compared to simulations using either the Emissions Database for Global Atmospheric Research (EDGAR) model or without

chemistry (bias approaching 100 %). The largest model-data mismatches are associated with substantial biases in wind directions or conditions of slower atmospheric mixing and photochemistry. More importantly, our model development illustrates (1) how NO_x chemistry affects the relationship between NO_x and CO₂ in terms of the spatial and seasonal variability and (2) how assimilating tNO₂ can quantify systematic biases in modeled wind directions and emission distribution in prior inventories of NO_x and CO₂, which laid a foundation for a local-scale multi-tracer emission optimization system.

3. The worldwide COVID-19 lockdown impacts on global secondary inorganic aerosols and radiative budget

Authors: Takashi Sekiya (Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan.); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan.); Henk Eskes (Royal Netherlands Meteorological Institute); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of California Los Angeles); Kengo Sudo (Nagoya University, Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan.); Yugo Kanaya (Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan.); Masayuki Takigawa (Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan.)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

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URL: <https://doi.org/10.1126/sciadv.adh2688>

Publication Date: 2023-07-28

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Cited by: 5 publications

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Open Access: N/A

Abstract: Global lockdown measures to prevent the spread of the coronavirus disease 2019 (COVID-19) led to air pollutant emission reductions. While the COVID-19 lockdown impacts on both trace gas and total particulate pollutants have been widely investigated, secondary aerosol formation from trace gases remains unclear. To that end, we quantify the COVID-19 lockdown impacts on NO_x and SO₂ emissions and sulfate-nitrate-ammonium aerosols using multiconstituent satellite data assimilation and model simulations. We find that anthropogenic emissions over major polluted regions were reduced by 19 to 25% for NO_x and 14 to 20% for SO₂ during April 2020. These emission reductions led to 8 to 21% decreases in sulfate and nitrate aerosols over highly polluted areas, corresponding to >34% of the observed aerosol optical depth declines and a global aerosol radiative forcing of +0.14 watts per square meter relative to business-as-usual scenario. These results point to the critical importance of secondary aerosol pollutants in quantifying climate impacts of future mitigation measures.

4. Impact of different sources of precursors on an ozone pollution outbreak over Europe analysed with IASI+GOME2 multispectral satellite observations and model simulations

Authors: Sachiko Okamoto; Juan Cuesta; Matthias Beekmann; Gaëlle Dufour; Maxim Eremenko; Kazuyuki Miyazaki; Cathy Boone; Hiroshi Tanimoto; Hajime Akimoto

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Cited by: 2 publications

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Keywords: N/A

Funding: Centre National d'Etudes Spatiales

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Open Access: N/A

Abstract: . We examine the impact of different sources of ozone precursors on the daily evolution of successive ozone pollution outbreaks across Europe in July 2017 by using a multispectral satellite approach called IASI+GOME2 and a tropospheric chemistry reanalysis named TCR-2. IASI+GOME2, combining IASI (Infrared Atmospheric Sounding Interferometer) and GOME-2 (Global Ozone Monitoring Experiment-2) measurements respectively in the infrared and the ultraviolet, allows the observation of the daily horizontal distribution of ozone in the lowermost troposphere (defined here as the atmospheric layer between the surface and 3 km above sea level). IASI+GOME2 observations show a fair capacity to depict near-surface ozone evolution as compared to surface measurements from 188 European stations for the period 15–27 July 2017. At the beginning of this event (on 16 July), an ozone outbreak is initially formed over the Iberian Peninsula likely linked with high temperature-induced enhancements of biogenic volatile organic compound concentrations and collocated anthropogenic emissions. In the following days, the ozone plume splits into two branches, one being transported eastward across the western Mediterranean and Italy and the other one over western and Central Europe. The southern branch encounters ozone precursor emitted over the Balkan Peninsula by wildfires along the coast of the Adriatic Sea and biogenic sources in the inland region of the peninsula. Ozone concentrations of the northern plume are enhanced by photochemical production associated with anthropogenic sources of ozone precursors over Central Europe and by mixing with an ozone plume arriving from the North Sea that was originally produced over North America. Finally, both ozone branches are transported eastwards and mix gradually as they reach the northern coast of the Black Sea. There, emissions from agricultural fires after harvesting clearly favour photochemical production of ozone within the pollution plume, which is advected eastwards in the following days. Based on satellite analysis, this paper shows the interplay of various ozone precursor sources to sustain a 2-week-long ozone pollution event over different parts of Europe.

5. Predictability of fossil fuel CO₂ from air quality emissions

Authors: Kazuyuki Miyazaki; Kevin Bowman

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-023-37264-8

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Language: en

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Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Quantifying the coevolution of greenhouse gases and air quality pollutants can provide insight into underlying anthropogenic processes enabling predictions of their emission trajectories. Here, we classify the dynamics of historic emissions in terms of a modified Environmental Kuznets Curve (MEKC), which postulates the coevolution of fossil fuel CO₂(FFCO₂) and NO_x emissions as a function of macroeconomic development. The MEKC broadly captures the historic FFCO₂-NO_xdynamical regimes for countries including the US, China, and India as well as IPCC scenarios. Given these dynamics, we find the predictive skill of FFCO₂ given NO_xemissions constrained by satellite data is less than 2% error at one-year lags for many countries and less than 10% for 4-year lags. The proposed framework in conjunction with an increasing satellite constellation provides valuable guidance to near-term emission scenario development and evaluation at time-scales relevant to international assessments such as the Global Stocktake.

6. Measurement report: Spatiotemporal variability of peroxy acyl nitrates (PANs) over Mexico City from TES and CrIS satellite measurements

Authors: Madison J. Shogrin; Vivienne H. Payne; Susan S. Kulawik; Kazuyuki Miyazaki; Emily V. Fischer

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-23-2667-2023

URL: <https://doi.org/10.5194/acp-23-2667-2023>

Publication Date: 2023-02-27

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References: 102

Cited by: 1 publications

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Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Peroxy acyl nitrates (PANs) are photochemical pollutants with implications for health and atmospheric oxidation capacity. PANs are formed via the oxidation of non-methane volatile organic compounds (NMVOCs) in the presence of nitrogen oxide radicals ($\text{NO}_x = \text{NO} + \text{NO}_2$). While urban environments are large sources of PANs, in situ observations in urban areas are limited. Here we use satellite measurements of PANs from the Tropospheric Emission Spectrometer (TES) and the Suomi National Polar-orbiting Partnership (S-NPP) Cross-track Infrared Sounder (CrIS) to evaluate the spatiotemporal variability of PANs over and around Mexico City. Monthly mean maxima in PANs over the Mexico City Metropolitan Area (MCMA) occur during spring months (March-May). This time of year coincides with a peak in local photochemistry and more frequent air stagnation. Local fire activity also typically peaks between February and May, which leads to strong interannual variability of PANs over the MCMA. We use S-NPP CrIS data to probe the spatial outflow pattern of PANs produced within urban Mexico City during the month with the largest mixing ratios of PANs (April). Peak outflow in April occurs to the northeast of the city and over the mountains south of the city. Outflow to the northwest appears infrequently. Using observations during 2018 versus 2019, we also show that PANs were not significantly reduced during a year, with a significant decrease in NO_x over Mexico City. Our analysis demonstrates that the space-based observations provided by CrIS and TES can increase understanding of the spatiotemporal variability of PANs over and surrounding Mexico City.

7. Optimizing the Isoprene Emission Model MEGAN With Satellite and Ground-Based Observational Constraints

Authors: Christian A. DiMaria (University of Toronto); Dylan B. A. Jones (University of Toronto); Helen Worden (National Center for Atmospheric Research); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Trissevgeni Stavrakou (Royal Belgian Institute for Space Aeronomy); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alex Guenther (University of California Irvine); Chinmoy Sarkar (University of California Irvine); Roger Seco (Institute of Environmental Assessment and Water Research (IDAEA-CSIC) Barcelona Spain); Jeong-Hoo Park (Air Quality Research Division National Institute of Environmental Research Incheon Republic of Korea); Julio Tota (Julio Tota Instituto de engenharia e Geociências Universidade Federal do Oeste do Pará UFOPA Santarém Brazil); Eliane Gomes Alves (Max Planck Institute for Biogeochemistry); Valerio Ferracci (Cranfield University)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: Isoprene is a hydrocarbon emitted in large quantities by terrestrial vegetation. It is a precursor to several air quality and climate pollutants including ozone. Emission rates vary with plant species and environmental conditions. This variability can be modeled using the Model of Emissions of Gases and Aerosols from Nature (MEGAN). MEGAN parameterizes isoprene emission rates as a vegetation-specific standard rate which is modulated by scaling factors that depend on meteorological and environmental driving variables. Recent experiments have identified large uncertainties in the MEGAN temperature response parameterization, while the emission rates under standard conditions are poorly constrained in some regions due to a lack of representative measurements and uncertainties in landcover. In this study, we use Bayesian model-data fusion to optimize the MEGAN temperature response and standard emission rates using satellite- and ground-based observational constraints. Optimization of the standard emission rate with satellite constraints reduced model biases but was highly sensitive to model input errors and drought stress and was found to be inconsistent with ground-based constraints at an Amazonian field site, reflecting large uncertainties in the satellite-based

emissions. Optimization of the temperature response with ground-based constraints increased the temperature sensitivity of the model by a factor of five at an Amazonian field site but had no impact at a UK field site, demonstrating significant ecosystem-dependent variability of the isoprene emission temperature sensitivity. Ground-based measurements of isoprene across a wide range of ecosystems will be key for obtaining an accurate representation of isoprene emission temperature sensitivity in global biogeochemical models.

8. Inverse modelling of Chinese NO_x emissions using deep learning: integrating in situ observations with a satellite-based chemical reanalysis

Authors: Tai-Long He; Dylan B. A. Jones; Kazuyuki Miyazaki; Kevin W. Bowman; Zhe Jiang; Xiaokang Chen; Rui Li; Yuxiang Zhang; Kunna Li

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Nitrogen dioxide (NO₂) column density measurements from satellites have been widely used in constraining emissions of nitrogen oxides (NO_x = NO + NO₂). However, the utility of these measurements is impacted by reduced observational coverage due to cloud cover and their reduced sensitivity toward the surface. Combining the information from satellites with surface observations of NO₂ will provide greater constraints on emission estimates of NO_x. We have developed a deep-learning (DL) model to integrate satellite data and in situ observations of surface NO₂ to estimate NO_x emissions in China. A priori information for the DL model was obtained from satellite-derived emissions from the Tropospheric Chemistry Reanalysis (TCR-2). A two-stage training strategy was used to integrate in situ measurements from the China Ministry of Ecology and Environment (MEE) observation network with the TCR-2 data. The DL model is trained from 2005 to 2018 and evaluated for 2019 and 2020. The DL model estimated a source of 19.4 Tg NO for total Chinese NO_x emissions in 2019, which is consistent with the TCR-2 estimate of 18.5 ± 3.9 Tg NO and the 20.9 Tg NO suggested by the Multi-resolution Emission Inventory for China (MEIC). Combining the MEE data with TCR-2, the DL model suggested higher NO_x emissions in some of the less-densely populated provinces, such as Shaanxi and Sichuan, where the MEE data indicated higher surface NO₂ concentrations than TCR-2. The DL model also suggested a faster recovery of NO_x emissions than TCR-2 after the Chinese New Year (CNY) holiday in 2019, with a recovery time scale that is consistent with Baidu "Qianxi" mobility data. In 2020, the DL-based analysis estimated about a 30 % reduction in NO_x emissions in eastern China during the COVID-19 lockdown period, relative to pre-lockdown levels. In particular, the maximum emission reductions were 42 % and 30 % for the Jing-Jin-Ji (JJJ) and the Yangtze River Delta (YRD) mega-regions, respectively. Our results illustrate the potential utility of the DL model as a complementary tool for conventional data-assimilation approaches for air quality applications.

9. Joint spectral retrievals of ozone with Suomi NPP CrIS augmented by S5P/TROPOMI

Authors: Edward Malina; Kevin W. Bowman; Valentin Kantchev; Le Kuai; Thomas P. Kurosu; Kazuyuki Miyazaki; Vijay Natraj; Gregory B. Osterman; Matthew D. Thill

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

Language: N/A

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Keywords: N/A

Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . The vertical distribution of ozone plays an important role in atmospheric chemistry, climate change, air pollution, and human health. Over the twenty-first century, spaceborne remote sensing methods and instrumentation have evolved to better characterise this distribution. We quantify the ability of ozone retrievals to characterise this distribution through a combination of thermal infrared (TIR) and Ultra Violet (UV) spectral radiances, harnessing co-located TIR measurements from the Cross Track Infrared Sounder (CrIS), onboard the Suomi National Polar-orbiting Partnership (NPP), and UV measurements from the TROPospheric Monitoring Instrument (TROPOMI), which is on the Sentinel 5-Precursor (S5P) satellite. The combination of TIR and UV measurements improves the ability of satellites to characterise global ozone profiles, over the use of each band individually. The CrIS retrievals enhanced by TROPOMI radiances in the Huggins band (325-335 nm) show good agreement with independent datasets both in the troposphere and in the stratosphere in spite of calibration issues in the TROPOMI UV. Improved performance is characterised in the stratosphere from CrIS-TROPOMI. Comparable performance between CrIS-TROPOMI and CrIS-only is found in the troposphere with degrees of freedom for signal of about 2 globally, but higher in the tropics partitioned equally between the lower and upper troposphere. These results demonstrate that CrIS/TROPOMI retrievals have the potential to substantially improve our understanding of ozone. If spectral accuracy is improved in future TROPOMI calibration, the degrees of freedom of signal in the stratosphere could double when using bands 1 and 2 of TROPOMI (270-330 nm), while tropospheric degrees of freedom of signal could increase by 25 %.

10. Satellite measurements of peroxyacetyl nitrate from the Cross-Track Infrared Sounder: comparison with ATom aircraft measurements

Authors: Vivienne H. Payne; Susan S. Kulawik; Emily V. Fischer; Jared F. Brewer; L. Gregory Huey; Kazuyuki Miyazaki; John R. Worden; Kevin W. Bowman; Eric J. Hintsa; Fred Moore; James W. Elkins; Julieta Juncosa Calahorrano

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

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Cited by: 5 publications

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Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . We present an overview of an optimal estimation algorithm to retrieve peroxyacetyl nitrate (PAN) from single-field-of-view Level 1B radiances measured by the Cross-Track Infrared Sounder (CrIS). CrIS PAN retrievals show peak sensitivity in the mid-troposphere, with degrees of freedom for signal less than or equal to 1.0. We show comparisons with two sets of aircraft measurements from the Atmospheric Tomography Mission (ATom), the PAN and Trace Hydrohalocarbon Experiment (PANTHER) and the Georgia Tech chemical ionization mass spectrometer (GT-CIMS). We find a systematic difference between the two aircraft datasets, with vertically averaged mid-tropospheric values from the GT-CIMS around 14 % lower than equivalent values from PANTHER. However, the two sets of aircraft measurements are strongly correlated (R^2 value of 0.92) and do provide a consistent view of the large-scale variation of PAN. We demonstrate that the retrievals of PAN from CrIS show skill in measurement of these large-scale PAN distributions in the remote mid-troposphere compared to the retrieval prior. The standard deviation of individual CrIS-aircraft differences is 0.08 ppbv, which we take as an estimate of the uncertainty of the CrIS mid-tropospheric PAN for a single satellite field of view. The standard deviation of the CrIS-aircraft comparisons for averaged CrIS retrievals (median of 20 satellite coincidences with each aircraft profile) is lower at 0.05 ppbv. This would suggest that the retrieval error is reduced with averaging, although not with the square root of the number of observations. We find a negative bias of the order of 0.1 ppbv in the CrIS PAN results with respect to the aircraft measurements. This bias shows a dependence on column water vapor. We provide a water-vapor-dependent bias correction for use with the CrIS PAN data.

11. Cause of a Lower-Tropospheric High-Ozone Layer in Spring Over Hanoi

Authors: S.-Y. Ogino (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan); K. Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); M. Fujiwara (Hokkaido University); M. I. Nodzu (Department of Geography Tokyo Metropolitan University Hachioji Japan); M. Shiotani (Kyoto University); F. Hasebe (Hokkaido University); J. Matsumoto (Japan Agency for Marine-Earth Science and Technology Yokosuka Japan, Department of Geography Tokyo Metropolitan University Hachioji Japan); J. Witte (National Center for Atmospheric Research); A. M. Thompson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); H. A. Nguyen-Thi (Viet Nam Meteorological and Hydrological Administration); T. V. Nguyen (Viet Nam Meteorological and Hydrological Administration)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

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References: 60

Cited by: 2 publications

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Keywords: N/A

Funding: Japan Society for the Promotion of Science

License: N/A

Open Access: N/A

Abstract: The ozonesonde observations in Hanoi, Vietnam, over 14 years since 2004, have confirmed an enhancement in lower tropospheric ozone concentration at about 3 km altitude in the spring season. We investigated the evolution of the ozone enhancement from analysis of meteorological data, backward trajectories, and model sensitivity experiments. In spring, air masses over Hanoi exhibit strong height dependence. At 3 km, the high-ozone air masses originate from the land area to the west of Hanoi, while low-ozone air masses below about 1.5 km are from the oceanic area to the east. Above 4 km, the air masses are mostly traced back to the farther west area. The chemical transport model simulations revealed that precursor emissions from biomass burning in the inland Indochina Peninsula have the largest contribution to the lower tropospheric ozone enhancement, which is transported upward and eastward and overhangs the clean air intrusion from the ocean to the east of Hanoi. At this height level, the polluted air has the horizontal extent of about 20 degrees in longitude and latitude. The polluted air observed in Hanoi is transported further east and widely spread over the northern Pacific Ocean.

12. 4D-Var Inversion of European NH₃ Emissions Using CrIS NH₃ Measurements and GEOS-Chem Adjoint With Bi-Directional and Uni-Directional Flux Schemes

Authors: Hansen Cao (University of Colorado Boulder); Daven K. Henze (University of Colorado Boulder); Liye Zhu (Sun Yat-Sen University); Mark W. Shephard (Environment and Climate Change Canada); Karen Cady-Pereira (Atmospheric and Environmental Research Inc.); Enrico Damers (Chinese Academy of Sciences); Michael Sitwell (Environment and Climate Change Canada); Nicholas Heath (Atmospheric and Environmental Research Inc.); Chantelle Lonsdale (Department of Civil, Structural and Environmental Engineering University at Buffalo Buffalo NY USA); Jesse O. Bash (U.S. Environmental Protection Agency Durham NC USA); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christophe Flechard (INRAE (National Research Institute for Agriculture, Food and Environment) UMR SAS Agrocampus Ouest Rennes France); Yannick Fauvel (INRAE (National Research Institute for Agriculture, Food and Environment) UMR SAS Agrocampus Ouest Rennes France); Roy Wichink Kruit (National Institute for Public Health and the Environment Bilthoven The Netherlands); Stefan Feigenspan (German Environment Agency Dessau-Roßlau Germany); Christian Brümmer (Thünen Institute of Climate-Smart Agriculture Braunschweig Germany); Frederik Schrader (Thünen Institute of Climate-Smart Agriculture Braunschweig Germany); Marsailidh M. Twigg (UK Centre for Ecology & Hydrology Penicuik UK); Sarah Leeson (UK Centre for Ecology & Hydrology Penicuik UK); Yuk S. Tang (UK Centre for Ecology & Hydrology Penicuik UK); Amy C. M. Stephens (UK Centre for Ecology & Hydrology Penicuik UK); Christine Braban (UK Centre for Ecology & Hydrology Penicuik UK); Keith Vincent (Ricardo Energy & Environment Wantage UK); Mario Meier (Forschungsstelle für Umweltbeobachtung Sankt Gallen Switzerland); Eva Seitler (Forschungsstelle für Umweltbeobachtung Sankt Gallen Switzerland); Camilla Geels (Aarhus University); Thomas Ellermann (Aarhus University); Agnieszka Sanocka (Ricardo Energy & Environment Wantage UK); Shannon L. Capps (Drexel University)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jd035687

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Publication Date: 2022-04-27

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Cited by: 8 publications

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Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: We conduct the first 4D-Var inversion of NH₃ accounting for NH₃ bi-directional flux, using CrIS satellite NH₃ observations over Europe in 2016. We find posterior NH₃ emissions peak more in

springtime than prior emissions at continental to national scales, and annually they are generally smaller than the prior emissions over central Europe, but larger over most of the rest of Europe. Annual posterior anthropogenic NH₃ emissions for 25 European Union members (EU25) are 25% higher than the prior emissions and very close (<2% difference) to other inventories. Our posterior annual anthropogenic emissions for EU25, the UK, the Netherlands, and Switzerland are generally 10%-20% smaller than when treating NH₃ fluxes as uni-directional emissions, while the monthly regional difference can be up to 34% (Switzerland in July). Compared to monthly mean in-situ observations, our posterior NH₃ emissions from both schemes generally improve the magnitude and seasonality of simulated surface NH₃ and bulk NH_x wet deposition throughout most of Europe, whereas evaluation against hourly measurements at a background site shows the bi-directional scheme better captures observed diurnal variability of surface NH₃. This contrast highlights the need for accurately simulating diurnal variability of NH₃ in assimilation of sun-synchronous observations and also the potential value of future geostationary satellite observations. Overall, our top-down ammonia emissions can help to examine the effectiveness of air pollution control policies to facilitate future air pollution management, as well as helping us understand the uncertainty in top-down NH₃ emissions estimates associated with treatment of NH₃ surface exchange.

13. Changes in US background ozone associated with the 2011 turnaround in Chinese NO_x emissions

Authors: Kazuyuki Miyazaki; Jessica L Neu; Greg Osterman; Kevin Bowman

Journal: Environmental Research Communications

Publisher: IOP Publishing

DOI: 10.1088/2515-7620/ac619b

URL: <https://doi.org/10.1088/2515-7620/ac619b>

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References: 72

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Language: N/A

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Keywords: N/A

Funding: NASA Atmospheric Composition: Aura Science Team Program, NASA Atmospheric Composition, NASA Health & Air Quality Applied Sciences Team

License: CC BY 4.0

Open Access: N/A

Abstract: Assessing the effectiveness of surface ozone air quality regulations is complicated by non-local sources and inhibited by sparse in situ observational networks. Here, we harness satellite measurements of ozone and its precursors over Asia to identify the Asian contribution to background ozone over the United States (US) from 2006-2016 using a state-of-the-art chemical data assimilation system. Our results indicate that Chinese NO_x emissions reached an apex in 2011, increasing daily average ozone concentrations over Los Angeles by up to 3.8 ppb at 750 hPa and 0.7 ppb at the surface in 2012 relative to 2006, with somewhat larger impacts in the high altitude regions of the Western US (assuming constant VOC emissions). Decreasing Chinese NO_x emissions after 2011 contributed to a reduction of similar magnitude in Western US background ozone from 2012-2016. We also assess the impact of direct assimilation of satellite-based tropospheric ozone profile measurements over Asia and find that it substantially modifies global ozone, including US surface concentrations. These results suggest that measurements of both ozone and its precursors over Asia play a substantial role in evaluating not only local but also global air quality. The simultaneous assimilation of ozone and its precursors proffers a powerful way to constrain the vertical profile of ozone and improve understanding of ozone variations. In the future, a more extensive satellite observing system has great potential to better constrain both local pollution and background ozone globally when employed in conjunction with chemical data assimilation.

14. A comparison of the impact of TROPOMI and OMI tropospheric NO₂ on global chemical data assimilation

Authors: Takashi Sekiya; Kazuyuki Miyazaki; Henk Eskes; Kengo Sudo; Masayuki Takigawa; Yugo Kanaya

Journal: Atmospheric Measurement Techniques

Publisher: Copernicus GmbH

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Funding: Ministry of Education, Culture, Sports, Science and Technology, Japan Society for the Promotion of Science

License: CC BY 4.0

Open Access: N/A

Abstract: . This study gives a systematic comparison of the Tropospheric Monitoring Instrument (TROPOMI) version 1.2 and Ozone Monitoring Instrument (OMI) QA4ECV tropospheric NO₂ column through global chemical data assimilation (DA) integration for the period April-May 2018. DA performance is controlled by measurement sensitivities, retrieval errors, and coverage. The smaller mean relative observation errors by 16 % in TROPOMI than OMI over 60° N-60° S during April-May 2018 led to larger reductions in the global root-mean-square error (RMSE) against the assimilated NO₂ measurements in TROPOMI DA (by 54 %) than in OMI DA (by 38 %). Agreements against the independent surface, aircraft-campaign, and ozonesonde observation data were also improved by TROPOMI DA compared to the control model simulation (by 12 %-84 % for NO₂ and by 7 %-40 % for ozone), which were more obvious than those by OMI DA for many cases (by 2 %-70 % for NO₂ and by 1 %-22 % for ozone) due to better capturing spatial and temporal variability by TROPOMI DA. The estimated global total NO_x emissions were 15 % lower in TROPOMI DA, with 2 %-23 % smaller regional total emissions, in line with the observed negative bias of the TROPOMI version 1.2 product compared to the OMI QA4ECV product. TROPOMI DA can provide city-scale emission estimates, which were within 10 % differences with other high-resolution analyses for several limited areas, while providing a globally consistent analysis. These results demonstrate that TROPOMI DA improves global analyses of NO₂ and ozone, which would also benefit studies on detailed spatial and temporal variations in ozone and nitrate aerosols and the evaluation of bottom-up NO_x emission inventories.

15. Deep Learning to Evaluate US NO_x Emissions Using Surface Ozone Predictions

Authors: Tai-Long He (University of Toronto); Dylan B. A. Jones (University of Toronto); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Binxuan Huang (Carnegie Mellon University); Yuyang Liu (University of Toronto); Zhe Jiang (University of Science and Technology of China); E. Charlie White (University of Toronto); Helen M. Worden (National Center for Atmospheric Research); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Atmospheres

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Keywords: N/A

Funding: Natural Sciences and Engineering Research Council of Canada, Canadian Space Agency, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Emissions of nitrogen oxides (NO_x = NO + NO₂) in the United States have declined significantly during the past three decades. However, satellite observations since 2009 indicate total column NO₂ is no longer declining even as bottom-up inventories suggest continued decline in emissions. Multiple explanations have been proposed for this discrepancy including (a) the increasing relative importance of nonurban NO_x to total column NO₂, (b) differences between background and urban NO_x lifetimes, and (c) that the actual NO_x emissions are declining more slowly after 2009. Here, we use a deep learning model trained by NO_x emissions and surface observations of ozone to assess consistency between the reported NO_x trends between 2005 and 2014 and observations of surface ozone. We find that the satellite-derived trends best reproduce ozone in low NO_x emission (background) regions. The 2010-2014 trend from older satellite-derived emission estimates produced at low spatial resolution results in the largest bias in surface ozone in regions with high NO_x emissions, reflecting the blending of urban and background NO_x in these low-resolution top-down analyses. In contrast, the trend from higher resolution satellite-based estimates, which are more capable of capturing the urban emission signature, is in better agreement with ozone in high NO_x emission regions, and is consistent with the trend based on surface observations of NO₂. Our results confirm that the satellite-derived trends reflect anthropogenic and background influences.

16. Large discrepancy between observed and modeled wintertime tropospheric NO₂ variabilities due to COVID-19 controls in China

Authors: Jiaqi Chen; Zhe Jiang; Rui Li; Chenggong Liao; Kazuyuki Miyazaki; Dylan B A Jones

Journal: Environmental Research Letters

Publisher: IOP Publishing

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URL: <https://doi.org/10.1088/1748-9326/ac4ec0>

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Abstract: Recent studies demonstrated the difficulties to explain observed tropospheric nitrogen dioxide (NO₂) variabilities over the United States and Europe, but thorough analysis for the impacts on tropospheric NO₂ in China is still lacking. Here we provide a comparative analysis for the observed and modeled (Goddard Earth Observing System-Chem) tropospheric NO₂ in early 2020 in China. Both ozone monitoring instrument and surface NO₂ measurements show marked decreases in NO₂ abundances due to the 2019 novel coronavirus (COVID-19) controls. However, we find a large discrepancy between observed and modeled NO₂ changes over highly polluted provinces: the observed reductions in tropospheric NO₂ columns are about 40% lower than those in surface NO₂ concentrations. By contrast, the modeled reductions in tropospheric NO₂ columns are about two times higher than those in surface NO₂ concentrations. This discrepancy could be driven by the combined effects from uncertainties in simulations and observations, associated with possible inaccurate simulations of lower tropospheric NO₂, larger uncertainties in the modeled interannual variabilities of NO₂ columns, as well as insufficient consideration of aerosol effects and a priori NO₂ variability in satellite retrievals. In addition, our analysis suggests a small influence from free tropospheric NO₂ backgrounds in E. China in winter. This work demonstrates the challenge to interpret wintertime tropospheric NO₂ changes in China, highlighting the importance of integrating surface NO₂ observations to provide better analysis for NO₂ variabilities.

17. Decadal Variabilities in Tropospheric Nitrogen Oxides Over United States, Europe, and China

Authors: Zhe Jiang (University of Science and Technology of China); Rui Zhu (University of Science and Technology of China); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Brian C. McDonald (NOAA Chemical Sciences Laboratory Boulder CO USA); Zbigniew Klimont (International Institute for Applied Systems Analysis (IIASA) Laxenburg Austria); Bo Zheng (Tsinghua University); K. Folkert Boersma (Wageningen University, Royal Netherlands Meteorological Institute); Qiang Zhang (Tsinghua University); Helen Worden (National Center for Atmospheric Research); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Daven K. Henze (University of Colorado Boulder); Dylan B. A. Jones (University of Toronto); Hugo A. C. Denier van der Gon (TNO Netherlands Organisation for Applied Scientific Research The Hague The Netherlands); Henk Eskes (Royal Netherlands Meteorological Institute)

Journal: Journal of Geophysical Research: Atmospheres

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Open Access: N/A

Abstract: Global trends in tropospheric nitrogen dioxide (NO₂) have changed dramatically in the past decade. Here, we investigate tropospheric NO₂ variabilities over United States, Europe, and E. China in 2005-2018 to explore the mechanisms governing the variation of this critical pollutant. We found large uncertainties in the trends of anthropogenic nitrogen oxides (NO_x) emissions, for example, the reductions of NO_x emissions, derived with different approaches and data sets, are in the range of 35%-50% over the United States and 15%-45% over Europe in 2005-2018. By contrast, the analysis in this work indicates declines of anthropogenic NO_x emissions by about 40% and 25% over the United States and Europe, respectively, in 2005-2018, and about 20% over E. China in 2012-2018. However, the shift of major NO_x sources from power generation to industrial and transportation sectors has led to noticeable diminishing effects in emission controls. Furthermore, satellite measurements exhibit the influence of NO₂ background levels over the United States and Europe, which offset the impacts of anthropogenic emission declines, resulting in flatter trends of tropospheric NO₂ over the United States and Europe. Our analysis further reveals underestimation of background NO₂ by chemical transport models, which can lead to inaccurate interpretations of satellite measurements. We use surface in-situ NO₂ observations to diagnose the satellite-observed NO₂ trends and find top-down NO_x emissions over urban grids represent the changes in anthropogenic NO_x emissions better. This work highlights

the importance of comprehensive applications of different analysis approaches to better characterizing atmospheric composition evolution.

18. The Carbon Cycle of Southeast Australia During 2019-2020: Drought, Fires, and Subsequent Recovery

Authors: B. Byrne (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Yin (California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); K. Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. J. Norton (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Joiner (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); D. F. Pollard (National Institute of Water and Atmospheric Research Ltd. (NIWA) Lauder New Zealand); D. W. T. Griffith (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); V. A. Velazco (Deutscher Wetterdienst German Meteorological Service Meteorological Observatory Hohenpeissenberg Germany, Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); N. M. Deutscher (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); N. B. Jones (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia); C. Paton-Walsh (Centre for Atmospheric Chemistry School of Earth, Atmospheric and Life Sciences University of Wollongong Wollongong NSW Australia)

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Open Access: N/A

Abstract: 2019 was the hottest and driest year on record for southeast Australia leading to bushfires of unprecedented extent. Ecosystem carbon losses due to drought and fire are believed to have been substantial, but have not been well quantified. Here, we utilize space-based measurements of trace gases (TROPOspheric Monitoring Instrument XCO, Orbiting Carbon Observatory 2) and up-scaled GPP (FluxSat GPP) to quantify the carbon cycle anomalies resulting from drought and fire in southeast Australia during the 2019-2020 growing season. We find that biomass burning released 113-236 TgC

of CO₂ while drought and fire-induced anomalies in net ecosystem exchange reduced growing season carbon uptake by an additional 19-52 TgC of CO₂. These carbon losses were concentrated during the spring and early summer, when hot-dry conditions were most severe. A shift to cooler conditions with above average rainfall during February is found to result in a partial recovery and greening in unburned ecosystems, but not in fire-impacted areas. The net 2019-2020 carbon loss substantially exceeded interannual variations in net uptake over 2010-2019 estimated from top-down constraints (~5sigma anomaly), and exceeded Australia's annual fossil fuel emissions (~104 TgC year⁻¹). Top-down constraints show that the regional carbon budget is strongly regulated by climate variability, and suggest that cool-wet conditions are required for a rapid recovery of carbon stocks. This has implications for the regional carbon budget as more frequent climate-change-driven heat and drought events may increase the frequency of fire events and the recovery time of ecosystems, threatening the carbon stocks of the region.

19. Societal shifts due to COVID-19 reveal large-scale complexities and feedbacks between atmospheric chemistry and climate change

Authors: Joshua L. Laughner (California Institute of Technology); Jessica L. Neu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Paul O. Wennberg (California Institute of Technology); Kelley Barsanti (University of California Riverside, Center for Environmental Research and Technology, Riverside, CA 92507;); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Abhishek Chatterjee (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Bart E. Croes (University of Colorado Boulder, Energy Research and Development Division, California Energy Commission, Sacramento, CA 95814;); Helen L. Fitzmaurice (University of California Berkeley); Daven K. Henze (University of Colorado Boulder); Jinsol Kim (University of California Berkeley); Eric A. Kort (University of Michigan); Zhu Liu (Tsinghua University); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexander J. Turner (University of Washington, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Berkeley); Susan Anenberg (George Washington University); Jeremy Aise (Modeling and Meteorology Branch, California Air Resources Board, Sacramento, CA 95814;); Hansen Cao (University of Colorado Boulder); David Crisp (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Joost de Gouw (University of Colorado Boulder); Annmarie Eldering (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John C. Fyfe (Canadian Centre for Climate Modelling and Analysis, Environment and Climate Change Canada); Daniel L. Goldberg (George Washington University); Kevin R. Gurney (Northern Arizona University); Sina Hasheminassab (South Coast Air Quality Management District); Francesca Hopkins (University of California Riverside); Cesunica E. Ivey (University of California Riverside, Center for Environmental Research and Technology, Riverside, CA 92507;); Dylan B. A. Jones (University of Toronto); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicole S. Lovenduski (University of Colorado Boulder); Randall V. Martin (Washington University in St. Louis); Galen A. McKinley (Columbia University); Lesley Ott (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Benjamin Poulter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Muye Ru (Duke University, Columbia University); Stanley P. Sander (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Neil Swart (Canadian Centre for Climate Modelling and Analysis, Environment and Climate Change Canada); Yuk L. Yung (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhao-Cheng Zeng (University of California Los Angeles); Unknown Author

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Keywords: N/A

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Abstract: SignificanceThe COVID-19 pandemic and associated lockdowns caused significant changes to human activity that temporarily altered our imprint on the atmosphere, providing a brief glimpse of potential future changes in atmospheric composition. This event demonstrated key feedbacks within and between air quality and the carbon cycle: Improvements in air quality increased the lifetime of methane (an important greenhouse gas), while unusually hot weather and intense wildfires in Los Angeles drove poor air quality. This shows that efforts to reduce greenhouse gas emissions and improve air quality cannot be considered separately.

20. Global tropospheric ozone responses to reduced NO_x emissions linked to the COVID-19 worldwide lockdowns

Authors: Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Takashi Sekiya (Japan Agency for Marine-Earth Science and Technology, Yokohama 236-0001, Japan.); Masayuki Takigawa (Japan Agency for Marine-Earth Science and Technology, Yokohama 236-0001, Japan.); Jessica L. Neu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kengo Sudo (Nagoya University, Japan Agency for Marine-Earth Science and Technology, Yokohama 236-0001, Japan.); Greg Osterman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Henk Eskes (Royal Netherlands Meteorological Institute)

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Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: Global ozone plunged during the COVID-19 pandemic: a glimpse into future benefits for air quality and climate.

21. Modeling the impact of COVID-19 on air quality in southern California: implications for future control policies

Authors: Zhe Jiang; Hongrong Shi; Bin Zhao; Yu Gu; Yifang Zhu; Kazuyuki Miyazaki; Xin Lu; Yuqiang Zhang; Kevin W. Bowman; Takashi Sekiya; Kuo-Nan Liou

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

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Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration, National Natural Science Foundation of China (and 4 more)

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Open Access: N/A

Abstract: . In response to the coronavirus disease of 2019 (COVID-19), California issued statewide stay-at-home orders, bringing about abrupt and dramatic reductions in air pollutant emissions. This crisis offers us an unprecedented opportunity to evaluate the effectiveness of emission reductions in terms of air quality. Here we use the Weather Research and Forecasting model with Chemistry (WRF-Chem) in combination with surface observations to study the impact of the COVID-19 lockdown measures on air quality in southern California. Based on activity level statistics and satellite observations, we estimate the sectoral emission changes during the lockdown. Due to the reduced emissions, the population-weighted concentrations of fine particulate matter (PM_{2.5}) decrease by 15 % in southern California. The emission reductions contribute 68 % of the PM_{2.5} concentration decrease before and after the lockdown, while meteorology variations contribute the remaining 32 %. Among all chemical compositions, the PM_{2.5} concentration decrease due to emission reductions is dominated by nitrate and primary components. For O₃ concentrations, the emission reductions cause a decrease in rural areas but an increase in urban areas; the increase can be offset by a 70 % emission reduction in anthropogenic volatile organic compounds (VOCs). These findings suggest that a strengthened control on primary PM_{2.5} emissions and a well-balanced control on nitrogen oxides and VOC emissions are needed to effectively and sustainably alleviate PM_{2.5} and O₃ pollution in southern California.

22. Impacts of Horizontal Resolution on Global Data Assimilation of Satellite Measurements for Tropospheric Chemistry Analysis

Authors: Takashi Sekiya (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Koji Ogochi (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Kengo Sudo (Nagoya University, Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Masayuki Takigawa (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Henk Eskes (Royal Netherlands Meteorological Institute); K. Folkert Boersma (Wageningen University, Royal Netherlands Meteorological Institute)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: We present the results from a global 0.56 degrees-resolution chemical data assimilation that integrates satellite observations of ozone, NO₂, CO, HNO₃, and SO₂ from OMI, GOME-2, SCIAMACHY, TES, MOPITT, and MLS. The assimilation is based on an ensemble Kalman filter technique and simultaneously optimizes ozone precursor emissions and concentrations of various species. The data assimilation at 0.56 degrees resolution reduced model errors against independent surface, aircraft, and ozonesonde observations, which was larger than at coarser resolutions for many cases. By the data assimilation, surface model errors over major polluted regions were reduced by 33%-75% for NO₂ and by 15%-18% for ozone. Agreements against assimilated observations for NO₂ were improved using the data assimilation at 0.56 degrees resolution by a factor of 1.5-3 compared to 2.8 degrees resolution over major polluted regions. The estimated global total NO_x emission over medium and strong source areas were smaller by 15% at 0.56 degrees resolution than at 2.8 degrees resolution associated with resolving small-scale transport and chemistry processes, while 2%-26% smaller emissions were found for regional total emissions over Europe, the United States, China, India, and South Africa, with larger differences over megacities such as Los Angeles (-41%). The estimated ship emissions were 5%-7% smaller at 0.56 degrees resolution over the Pacific and Atlantic. The 0.56 degrees-resolution data assimilation provides globally consistent analyses of the emissions and concentrations on a megacity scale, which benefit studies on air quality and its impact on human health at various spatial scales over different regions of the world.

23. Co-evolution of carbon cycle and air quality fluxes constrained by CMS-Flux and MOMO-Chem assimilation systems

Authors: Kevin Bowman; Kazuyuki Miyazaki; Junjie Liu; Anthony Bloom

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Carbon flux|Carbon fibers|Atmospheric carbon cycle

Funding: N/A

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Abstract: <p>Rapid regional changes in anthropogenic emissions in response to the COVID-19 pandemic have underscored the contribution of fossil fuel (FF) emission uncertainty to regional carbon budgets. Typical methods for spatially-explicit FF emissions are dependent on national reporting, which can incur substantial latencies. However, the concomitant changes in short-lived pollutants from common emission sources point to opportunities to develop independent low-latency estimates of fossil fuel emissions and to better understand anthropogenic processes. Here we combine state-of-the-art Multiple Model Multi Constituent chemical data assimilation system (MOMO-Chem) with bottom-up FF emissions to repartition the net carbon fluxes from the NASA Carbon Monitoring System Flux (CMS-Flux) project. To that end, we implement a novel Kalman filtering algorithm that predicts emission ratio co-evolution of air quality (AQ) and carbon species. Based upon top-down estimates of AQ emissions, FF CO₂ emissions and uncertainties can be rapidly determined. We show overall good agreement between predicted FF fluxes and the latest bottom-up inventories. These data are in turn used to interpret the decadal evolution of CMS-Flux net carbon exchange. This approach is an important step in quantifying both regional fossil fuel and natural carbon fluxes contributions to the atmospheric CO₂ growth rate.</p>

24. Correcting model biases of CO in East Asia: impact on oxidant distributions during KORUS-AQ

Authors: Benjamin Gaubert; Louisa K. Emmons; Kevin Raeder; Simone Tilmes; Kazuyuki Miyazaki; Avelino F. Arellano Jr.; Nellie Elguindi; Claire Granier; Wenfu Tang; Jérôme Barré; Helen M. Worden; Rebecca R. Buchholz; David P. Edwards; Philipp Franke; Jeffrey L. Anderson; Marielle Saunoy; Jason Schroeder; Jung-Hun Woo; Isobel J. Simpson; Donald R. Blake; Simone Meinardi; Paul O. Wennberg; John Crounse; Alex Teng; Michelle Kim; Russell R. Dickerson; Hao He; Xinrong Ren; Sally E. Pusede; Glenn S. Diskin

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Funding: National Aeronautics and Space Administration, National Oceanic and Atmospheric Administration

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Abstract: . Global coupled chemistry-climate models underestimate carbonmonoxide (CO) in the Northern Hemisphere, exhibiting a pervasive negativebias against measurements peaking in late winter and early spring. Whilethis bias has been commonly attributed to underestimation of directanthropogenic and biomass burning emissions, chemical production and loss via OH reaction from emissions of anthropogenic and biogenic volatile organic compounds (VOCs) play animportant role. Here we investigate the reasons for this underestimation using aircraft measurements taken in May and June 2016 from the Korea-United States Air Quality (KORUS-AQ) experiment in South Korea and the AirChemistry Research in Asia (ARIAs) in the North China Plain (NCP). Forreference, multispectral CO retrievals (V8J) from the Measurements of Pollution in the Troposphere (MOPITT) are jointly assimilated with meteorological observations using an ensemble adjustment Kalman filter (EAKF) within the global Community Atmosphere Model with Chemistry (CAM-Chem) and the Data Assimilation Research Testbed (DART). With regard to KORUS-AQ data, CO is underestimated by 42 % in the control run and by 12 % with the MOPITT assimilation run. The inversion suggests an underestimation of anthropogenic CO sources in many regions, by up to 80 % for northern China, with large increments over the Liaoning Province and the North China Plain (NCP). Yet, an often-overlooked aspect of these inversions is that correcting the underestimation in anthropogenic CO emissions also improves the comparison with observational O₃ datasets and observationally constrained box model simulations of OH and HO₂. Running a CAM-Chem simulation with the updated emissions of anthropogenic CO reduces the bias by 29 % for CO, 18 % for ozone, 11 % for HO₂, and 27 % for OH. Longer-lived anthropogenic VOCs whose model errors are correlated with CO are also improved, while short-lived VOCs, including

formaldehyde, are difficult to constrain solely by assimilating satellite retrievals of CO. During an anticyclonic episode, better simulation of O₃, with an average underestimation of 5.5 ppbv, and a reduction in the bias of surface formaldehyde and oxygenated VOCs can be achieved by separately increasing by a factor of 2 the modeled biogenic emissions for the plant functional types found in Korea. Results also suggest that controlling VOC and CO emissions, in addition to widespread NO_x controls, can improve ozone pollution over East Asia.

25. Long-term evaluation of atmospheric composition reanalyses from CAMS, TCR-2, and MERRA-2 over South Korea: Insights into applications, implications, and limitations

Authors: Young-Hee Ryu; Seung-Ki Min

Journal: Atmospheric Environment

Publisher: Elsevier BV

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Abstract: N/A

26. Air Quality Response in China Linked to the 2019 Novel Coronavirus (COVID-19) Lockdown

Authors: K. Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); K. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. Sekiya (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Z. Jiang (University of Science and Technology of China); X. Chen (University of Science and Technology of China); H. Eskes (Royal Netherlands Meteorological Institute); M. Ru (Duke University); Y. Zhang (Duke University); D. Shindell (Duke University, Tel Aviv University)

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Abstract: Efforts to stem the spread of COVID-19 in China hinged on severe restrictions to human movement starting 23 January 2020 in Wuhan and subsequently to other provinces. Here, we quantify the ancillary impacts on air pollution and human health using inverse emissions estimates based on multiple satellite observations. We find that Chinese NO_x emissions were reduced by 36% from early January to mid-February, with more than 80% of reductions occurring after their respective lockdown in most provinces. The reduced precursor emissions increased surface ozone by up to 16 ppb over northern China but decreased PM_{2.5} by up to 23 µg m⁻³ nationwide. Changes in human exposure are associated with about 2,100 more ozone-related and at least 60,000 fewer PM_{2.5}-related morbidity incidences, primarily from asthma cases, thereby augmenting efforts to reduce hospital admissions and alleviate negative impacts from potential delayed treatments.

27. Updated tropospheric chemistry reanalysis and emission estimates, TCR-2, for 2005-2018

Authors: Kazuyuki Miyazaki; Kevin Bowman; Takashi Sekiya; Henk Eskes; Folkert Boersma; Helen Worden; Nathaniel Livesey; Vivienne H. Payne; Kengo Sudo; Yugo Kanaya; Masayuki Takigawa; Koji Ogochi

Journal: Earth System Science Data

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Abstract: . This study presents the results from the Tropospheric Chemistry Reanalysis version 2 (TCR-2) for the period 2005-2018 at 1.1° horizontal resolution obtained from the assimilation of multiple updated satellite measurements of ozone, CO, NO₂, HNO₃, and SO₂ from the OMI, SCIAMACHY, GOME-2, TES, MLS, and MOPITT satellite instruments. The reanalysis calculation was conducted using a global chemical transport model MIROC-CHASER and an ensemble Kalman filter technique that optimizes both chemical concentrations of various species and emissions of several precursors, which was efficient for the correction of the entire tropospheric profile of various species and its year-to-year variations. Comparisons against independent aircraft, satellite, and ozonesonde observations demonstrate the quality of the reanalysis fields for numerous key species on regional and global scales, as well as for seasonal, yearly, and decadal scales, from the surface to the lower stratosphere. The multi-constituent data assimilation brought the model vertical profiles and interhemispheric gradient of OH closer to observational estimates, which was important in improving the description of the oxidation capacity of the atmosphere and thus vertical profiles of various species. The evaluation results demonstrate the capability of the chemical reanalysis to improve understanding of the processes controlling variations in atmospheric composition, including long-term changes in near-surface air quality and emissions. The estimated emissions can be employed for the elucidation of detailed distributions of the anthropogenic and biomass burning emissions of co-emitted species (NO_x, CO, SO₂) in all major regions, as well as their seasonal and decadal variabilities. The data sets are available at <https://doi.org/10.25966/9qgv-fe81> (Miyazaki et al., 2019a).

28. Evaluation of tropospheric ozone reanalyses with independent ozonesonde observations in East Asia

Authors: Sunmin Park; Seok-Woo Son; Myung-II Jung; Jinkyung Park; Sang Seo Park

Journal: Geoscience Letters

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Abstract: The modern reanalysis datasets provide not only meteorological variables, but also atmospheric chemical compositions such as tropospheric ozone and aerosol concentration. However, the quality of chemical compositions has been rarely assessed especially over East Asia. To better understand the characteristics of reanalysis datasets on regional scale, the present study evaluates tropospheric ozone derived from seven reanalyses against five independent ozonesonde observations in East Asia. The reanalysis datasets are the ECMWF Reanalysis 5th (ERA5), Monitoring Atmospheric Composition and Climate reanalysis (MACCRA), Copernicus Atmosphere Monitoring Service reanalysis (CAMSRA), as well as the NCEP Climate Forecast System Reanalysis (CFSR), NASA Modern-Era Retrospective analysis for Research and Applications version 2 (MERRA2), Japanese 55-year Reanalysis (JRA-55), and updated Tropospheric Chemistry Reanalysis (TCR-2). It turns out that MACCRA, CAMSRA, and TCR-2, which incorporate chemical transport model, depict most reasonable spatio-temporal variability of tropospheric ozone in East Asia. The MACC exhibits a better quality with relatively small mean biases of 6.4 +/- 1.3% in tropospheric column ozone than biases of 7.8 +/- 2.7% and 7.8 +/- 2.8% for CAMSRA and TCR-2. The CAMSRA further shows a significant monthly correlation with the observation of up to 0.7 at 850 hPa. Among the seven reanalyses, MACC, CAMSRA, and TCR-2 are suitable for local tropospheric ozone study on seasonal to inter-annual time scales. However, none of the seven reanalysis datasets reproduce the observed trend of tropospheric ozone. This result suggests that even the latest datasets are inadequate for the long-term ozone change study.

29. Intercomparison of Magnitudes and Trends in Anthropogenic Surface Emissions From Bottom-Up Inventories, Top-Down Estimates, and Emission Scenarios

Authors: N. Elguindi (Laboratoire d'Aérodynamique, CNRS Université de Toulouse Toulouse France); C. Granier (Laboratoire d'Aérodynamique, CNRS Université de Toulouse Toulouse France, University of Colorado Boulder); T. Stavrakou (Royal Belgian Institute for Space Aeronomy); S. Darras (Observatoire Midi-Pyrénées Toulouse France); M. Bauwens (Royal Belgian Institute for Space Aeronomy); H. Cao (University of Colorado Boulder); C. Chen (Laboratoire d'Optique Atmosphérique (LOA), UMR8518 CNRS Université de Lille Lille France, GRASP-SAS, Remote Sensing Developments Université de Lille Lille France); H. A. C. Denier van der Gon (Department of Climate, Air and Sustainability TNO Utrecht The Netherlands); O. Dubovik (Laboratoire d'Optique Atmosphérique (LOA), UMR8518 CNRS Université de Lille Lille France); T. M. Fu (Southern University of Science and Technology); D. K. Henze (University of Colorado Boulder); Z. Jiang (University of Science and Technology of China); S. Keita (Laboratoire de Physique de l'Atmosphère (LAPA-UFHB) University PGC Korhogo Côte d'Ivoire, UFR sciences Biologiques Université PGC Korhogo Côte d'Ivoire); J. J. P. Kuenen (Department of Climate, Air and Sustainability TNO Utrecht The Netherlands); J. Kurokawa (Asia Center for Air Pollution Research Niigata Japan); C. Lioussé (Laboratoire d'Aérodynamique, CNRS Université de Toulouse Toulouse France); K. Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J.-F. Müller (Royal Belgian Institute for Space Aeronomy); Z. Qu (Harvard University, University of Colorado Boulder); F. Solmon (Laboratoire d'Aérodynamique, CNRS Université de Toulouse Toulouse France); B. Zheng (Laboratoire des Sciences du Climat et de l'Environnement CEA-CNRS-UVSQ, UMR8212 Gif-sur-Yvette France)

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Abstract: This study compares recent CO, NO_x, NMVOC, SO₂, BC, and OC anthropogenic emissions from several state-of-the-art top-down estimates to global and regional bottom-up inventories and projections from five Shared Socioeconomic Pathways (SSPs) in several regions. Results show that top-down emissions derived in several recent studies exhibit similar uncertainty as bottom-up inventories in some regions for certain species and even less in the case of Chinese CO emissions. In general, the largest discrepancies are found outside of regions such as the United States, Europe, and

Japan where the most accurate and detailed information on emissions is available. In some regions such as China, which has recently undergone dynamical economic growth and changes in air quality regulations, the top-down estimates better capture recent emission trends than global bottom-up inventories. These results show the potential of top-down estimates to complement bottom-up inventories and to aide in the development of emission scenarios, particularly in regions where global inventories lack the necessary up-to-date and accurate information regarding regional activity data and emission factors such as Africa and India. Areas of future work aimed at quantifying and reducing uncertainty are also highlighted. A regional comparison of recent CO and NO_x trends in the five SSPs indicate that SSP126, a strong pollution control scenario, best represents the trends from the top-down and regional bottom-up inventories in the United States, Europe, and China, while SSP460, a low-pollution control scenario, lies closest to actual trends in West Africa. This analysis can be useful for air quality forecasting and near-future pollution control/mitigation policy studies.

30. An intercomparison of tropospheric ozone reanalysis products from CAMS, CAMS interim, TCR-1, and TCR-2

Authors: Vincent Huijnen; Kazuyuki Miyazaki; Johannes Flemming; Antje Inness; Takashi Sekiya; Martin G. Schultz

Journal: Geoscientific Model Development

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Abstract: . Global tropospheric ozone reanalyses constructed using different state-of-the-art satellite data assimilation systems, prepared as part of the Copernicus Atmosphere Monitoring Service (CAMS-iRean and CAMS-Rean) as well as two fully independent reanalyses (TCR-1 and TCR-2, Tropospheric Chemistry Reanalysis), have been intercompared and evaluated for the past decade. The updated reanalyses (CAMS-Rean and TCR-2) generally show substantially improved agreements with independent ground and ozone-sonde observations over their predecessor versions (CAMS-iRean and TCR-1) for diurnal, synoptical, seasonal, and interannual variabilities. For instance, for the Northern Hemisphere (NH) mid-latitudes the tropospheric ozone columns (surface to 300 hPa) from the updated reanalyses show mean biases to within 0.8 DU (Dobson units, 3 % relative to the observed column) with respect to the ozone-sonde observations. The improved performance can likely be attributed to a mixture of various upgrades, such as revisions in the chemical data assimilation, including the assimilated measurements, and the forecast model performance. The updated chemical reanalyses agree well with each other for most cases, which highlights the usefulness of the current chemical reanalyses in a variety of studies. Meanwhile, significant temporal changes in the reanalysis quality in all the systems can be attributed to discontinuities in the observing systems. To improve the temporal consistency, a careful assessment of changes in the assimilation configuration, such as a detailed assessment of biases between various retrieval products, is needed. Our comparison suggests that improving the observational constraints, including the continued development of satellite observing systems, together with the optimization of model parameterizations such as deposition and chemical reactions, will lead to increasingly consistent long-term reanalyses in the future.

31. Evaluation of a multi-model, multi-constituent assimilation framework for tropospheric chemical reanalysis

Authors: Kazuyuki Miyazaki; Kevin W. Bowman; Keiya Yumimoto; Thomas Walker; Kengo Sudo

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References: 105

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Abstract: . We introduce a Multi-mOdel Multi-cOnstituent Chemicaldata assimilation (MOMO-Chem) framework that directly accounts for modelerror in transport and chemistry, and we integrate a portfolio of dataassimilation analyses obtained using multiple forward chemical transportmodels in a state-of-the-art ensemble Kalman filter data assimilationssystem. The data assimilation simultaneously optimizes both concentrationsand emissions of multiple species through ingestion of a suite ofmeasurements (ozone, NO₂, CO, HNO₃) from multiple satellitesensors. In spite of substantial model differences, the observationaldensity and accuracy was sufficient for the assimilation to reduce themulti-model spread by 20 %-85 % for ozone and annual mean bias by39 %-97 % for ozone in the middle troposphere, while simultaneouslyreducing the tropospheric NO₂ column biases by more than 40 % andthe negative biases of surface CO in the Northern Hemisphere by 41 %-94 %.For tropospheric mean OH, the multi-model mean meridional hemisphericgradient was reduced from 1.32+/-0.03 to 1.19+/-0.03, while themulti-model spread was reduced by 24 %-58 % over polluted areas. Theuncertainty ranges in the a posteriori emissions due to model errors werequantified in 4 %-31 % for NO_x and 13 %-35 % for CO regional emissions.Harnessing assimilation increments in both NO_x and ozone, we show that thesensitivity of ozone and NO₂ surface concentrations to NO_x emissionsvaried by a factor of 2 for end-member models, revealing fundamentaldifferences in the representation of fast chemical and dynamical processes.A systematic investigation of model ozone response and analysis increment inMOMO-Chem could benefit evaluation of future prediction of the chemistry-climatesystem as a hierarchical emergent constraint.

32. Attribution of Chemistry-Climate Model Initiative (CCMI) ozone radiative flux bias from satellites

Authors: Le Kuai; Kevin W. Bowman; Kazuyuki Miyazaki; Makoto Deushi; Laura Revell; Eugene Rozanov; Fabien Paulot; Sarah Strode; Andrew Conley; Jean-François Lamarque; Patrick Jöckel; David A. Plummer; Luke D. Oman; Helen Worden; Susan Kulawik; David Paynter; Andrea Stenke; Markus Kunze

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Abstract: . The top-of-atmosphere (TOA) outgoing longwave flux over the 9.6 μm ozone band is a fundamental quantity for understanding chemistry-climate coupling. However, observed TOA fluxes are hard to estimate as they exhibit considerable variability in space and time that depend on the distribution of clouds, ozone (O_3), water vapor (H_2O), air temperature (T_a), and surface temperature (T_s). Benchmarking present-day fluxes and quantifying the relative influence of their drivers is the first step for estimating climate feedbacks from ozone radiative forcing and predicting radiative forcing evolution. To that end, we constructed observational instantaneous radiative kernels (IRKs) under clear-sky conditions, representing the sensitivities of the TOA flux in the 9.6 μm ozone band to the vertical distribution of geophysical variables, including O_3 , H_2O , T_a , and T_s based upon the Aura Tropospheric Emission Spectrometer (TES) measurements. Applying these kernels to present-day simulations from the Chemistry-Climate Model Initiative (CCMI) project as compared to a 2006 reanalysis assimilating satellite observations, we show that the models have large differences in TOA flux, attributable to different geophysical variables. In particular, model simulations continue to diverge from observations in the tropics, as reported in previous studies of the Atmospheric Chemistry Climate Model Intercomparison Project (ACCMIP) simulations. The principal culprits are tropical middle and upper tropospheric ozone followed by tropical lower tropospheric H_2O . Five models out of the eight studied here have TOA flux biases exceeding 100 mW m^{-2} attributable to tropospheric ozone bias. Another set of five models have flux biases over 50 mW m^{-2} due to H_2O . On the other hand, T_a radiative bias is negligible in all models (no more than 30 mW m^{-2}). We found that the atmospheric component (AM3) of the Geophysical Fluid Dynamics Laboratory (GFDL) general circulation model and Canadian Middle Atmosphere Model (CMAM) have the lowest TOA flux biases globally but are a result of cancellation of opposite biases due to different processes. Overall, the multi-model ensemble mean bias is $-133 \pm 98 \text{ mW m}^{-2}$, indicating that they are too atmospherically opaque due to trapping too

much radiation in the atmosphere by overestimated tropical tropospheric O₃ and H₂O. Having too much O₃ and H₂O in the troposphere would have different impacts on the sensitivity of TOA flux to O₃ and these competing effects add more uncertainties on the ozone radiative forcing. We find that the inter-model TOA outgoing longwave radiation (OLR) difference is well anti-correlated with their ozone band flux bias. This suggests that there is significant radiative compensation in the calculation of model outgoing longwave radiation.

33. Comparison of Near-Surface NO₂ Pollution With Pandora Total Column NO₂ During the Korea-United States Ocean Color (KORUS OC) Campaign

Authors: Anne M. Thompson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Ryan M. Stauffer (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, USRA Columbia MD USA); Tyler P. Boyle (University of Maryland); Debra E. Kollonige (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, SSAI Lanham MD USA); Kazuyuki Miyazaki (Jet Propulsion Lab Cal Tech Pasadena CA USA); Maria Tzortziou (Earth Sciences Department CCNY New York NY USA); Jay R. Herman (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Nader Abuhassan (University of Maryland Baltimore County, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Carolyn E. Jordan (National Institute of Aerospace Hampton VA USA); Brian T. Lamb (Earth Sciences Department CCNY New York NY USA)

Journal: Journal of Geophysical Research: Atmospheres

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Abstract: Near-surface air quality (AQ) observations over coastal waters are scarce, a situation that limits our capacity to monitor pollution events at land-water interfaces. Satellite measurements of total column (TC) nitrogen dioxide (NO₂) observations are a useful proxy for combustion sources, but the once daily snapshots available from most sensors are insufficient for tracking the diurnal evolution and transport of pollution. Ground-based remote sensors like the Pandora Spectrometer Instrument (PSI) that have been developed to verify space-based TC NO₂ and other trace gases are being tested for routine use as certified AQ monitors. The KORUS-OC (Korea-United States Ocean Color) cruise aboard the R/V Onnuri in May-June 2016 represented an opportunity to study AQ near the South Korean coast, a region affected by both local/regional and long-distance pollution sources. Using PSI data in direct-Sun mode and in situ sensors for shipboard ozone, CO, and NO₂, we explore, for the first time, relationships between TC NO₂ and surface AQ in this coastal region. Three case studies illustrate the value of the PSI and complexities in the surface-column NO₂ relationship caused by varying meteorological conditions. Case Study 1 (25-26 May 2016) exhibited a high correlation of surface NO₂ to TC NO₂ measured by both PSI and Aura's Ozone Monitoring Instrument, but two other cases displayed poor relationships between in situ and TC NO₂ due to decoupling of pollution layers from the

surface. With suitable interpretation the PSI TC NO₂ measurement demonstrates good potential for working with upcoming geostationary satellites to advance diurnal tracking of pollution.

34. Ozone and carbon monoxide observations over open oceans on R/V Mirai from 67 degrees S to 75 degrees N during 2012 to 2017: testing global chemical reanalysis in terms of Arctic processes, low ozone levels at low latitudes, and pollution transport

Authors: Yugo Kanaya; Kazuyuki Miyazaki; Fumikazu Taketani; Takuma Miyakawa; Hisahiro Takashima; Yuichi Komazaki; Xiaole Pan; Saki Kato; Kengo Sudo; Takashi Sekiya; Jun Inoue; Kazutoshi Sato; Kazuhiro Oshima

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Abstract: . Constraints from ozone (O₃) observations over oceans are needed in addition to those from terrestrial regions to fully understand global tropospheric chemistry and its impact on the climate. Here, we provide a large data set of ozone and carbon monoxide (CO) levels observed (for 11 666 and 10 681 h, respectively) over oceans. The data set is derived from observations made during 24 research cruise legs of R/V Mirai during 2012 to 2017, in the Southern, Indian, Pacific, and Arctic oceans, covering the region from 67° S to 75° N. The data are suitable for critical evaluation of the over-ocean distribution of ozone derived from global atmospheric chemistry models. We first give an overview of the statistics in the data set and highlight key features in terms of geographical distribution and air mass type. We then use the data set to evaluate ozone mixing ratio fields from the tropospheric chemistry reanalysis version 2 (TCR-2), produced by assimilating a suite of satellite observations of multiple species into a global atmospheric chemistry model, namely CHASER. For long-range transport of polluted air masses from continents to the oceans, during which the effects of forest fires and fossil fuel combustion were recognized, TCR-2 gave an excellent performance in reproducing the observed temporal variations and photochemical buildup of O₃ when assessed from $\Delta\text{O}_3/\Delta\text{CO}$ ratios. For clean marine conditions with low and stable CO mixing ratios, two focused analyses were performed. The first was in the Arctic (> 70° N) in September every year from 2013 to 2016; TCR-2 underpredicted O₃ levels by 6.7 ppbv (21 %) on average. The observed vertical profiles from O₃ soundings from R/V Mirai during September 2014 had less steep vertical gradients at low altitudes (> 850 hPa) than those obtained by TCR-2. This suggests the possibility of a more efficient descent of the O₃-rich air from above than assumed in the models. For TCR-2 (CHASER), dry deposition on the Arctic ocean surface might also have been overestimated. In the second analysis, over the western Pacific equatorial region

(125-165° E, 10° S to 25° N), the observed O₃ level more frequently decreased to less than 10 ppbv in comparison to that obtained with TCR-2 and also those obtained in most of the Atmospheric Chemistry Climate Model Intercomparison Project (ACCMIP) model runs for the decade from 2000. These results imply loss processes that are unaccounted for in the models. We found that the model's positive bias positively correlated with the daytime residence times of air masses over a particular grid, namely 165-180° E and 15-30° N; an additional loss rate of 0.25 ppbv h⁻¹ in the grid best explained the gap. Halogen chemistry, which is commonly omitted from currently used models, might be active in this region and could have contributed to additional losses. Our open data set covering wide ocean regions is complementary to the Tropospheric Ozone Assessment Report data set, which basically comprises ground-based observations and enables a fully global study of the behavior of O₃.

35. Satellite data reveal a common combustion emission pathway for major cities in China

Authors: Wenfu Tang; Avelino F. Arellano; Benjamin Gaubert; Kazuyuki Miyazaki; Helen M. Worden

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Publisher: Copernicus GmbH

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Abstract: . Extensive fossil fuel combustion in rapidly developing cities severely affects air quality and public health. We report observational evidence of decadal changes in the efficiency and cleanliness of bulk combustion over large cities in mainland China. In order to estimate the trends in enhancement ratios of CO and SO₂ to NO₂ ($\Delta\text{CO}/\Delta\text{NO}_2$ and $\Delta\text{SO}_2/\Delta\text{NO}_2$) and infer emergent bulk combustion properties over these cities, we combine air quality retrievals from widely used satellite instruments across 2005–2014. We present results for four Chinese cities (Shenyang, Beijing, Shanghai, and Shenzhen) representing four levels of urban development. Our results show a robust coherent progression of declining to growing $\Delta\text{CO}/\Delta\text{NO}_2$ relative to 2005 (-5.4 ± 0.7 to $+8.3 \pm 3.1\%$ yr⁻¹) and slowly declining $\Delta\text{SO}_2/\Delta\text{NO}_2$ (-6.0 ± 1.0 to $-3.4 \pm 1.0\%$ yr⁻¹) across the four cities. The coherent progression we find is not evident in the trends of emission ratios reported in Representative Concentration Pathway (RCP8.5) inventory. This progression is likely due to a shift towards cleaner combustion from industrial and residential sectors in Shanghai and Shenzhen that is not yet seen in Shenyang and Beijing. This overall trend is presently obscured by China's still relatively higher dependence on coal. Such progression is well-correlated with economic development and traces a common emission pathway that resembles evolution of air pollution in more developed cities. Our results highlight the utility of augmenting observing and modeling capabilities by exploiting enhancement ratios in constraining the time variation in emission ratios in current inventories. As cities and/or countries continue to socioeconomically develop, the ability to monitor combustion efficiency and effectiveness of pollution control becomes increasingly important in assessing sustainable control strategies.

36. Balance of Emission and Dynamical Controls on Ozone During the Korea-United States Air Quality Campaign From Multiconstituent Satellite Data Assimilation

Authors: K. Miyazaki (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); T. Sekiya (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); D. Fu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); K. W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. S. Kulawik (Bay Area Environmental Research Institute Sonoma CA USA); K. Sudo (Nagoya University, Japan Agency for Marine-Earth Science and Technology Yokohama Japan); T. Walker (Department of Civil and Environmental Engineering Carleton University Ottawa Ontario Canada); Y. Kanaya (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); M. Takigawa (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); K. Ogochi (Japan Agency for Marine-Earth Science and Technology Yokohama Japan); H. Eskes (Royal Netherlands Meteorological Institute); K. F. Boersma (Wageningen University, Royal Netherlands Meteorological Institute); A. M. Thompson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); B. Gaubert (National Center for Atmospheric Research); J. Barre (European Centre for Medium-Range Weather Forecasts (ECMWF)); L. K. Emmons (National Center for Atmospheric Research)

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Open Access: N/A

Abstract: Global multiconstituent concentration and emission fields obtained from the assimilation of the satellite retrievals of ozone, CO, NO₂, HNO₃, and SO₂ from the Ozone Monitoring Instrument (OMI), Global Ozone Monitoring Experiment 2, Measurements of Pollution in the Troposphere, Microwave Limb Sounder, and Atmospheric Infrared Sounder (AIRS)/OMI are used to understand the processes controlling air pollution during the Korea-United States Air Quality (KORUS-AQ) campaign. Estimated emissions in South Korea were 0.42 Tg N for NO_x and 1.1 Tg CO for CO, which were 40% and 83% higher, respectively, than the a priori bottom-up inventories, and increased mean ozone concentration by up to 7.5 +/- 1.6 ppbv. The observed boundary layer ozone exceeded 90 ppbv over Seoul under stagnant phases, whereas it was approximately 60 ppbv during dynamical conditions

given equivalent emissions. Chemical reanalysis showed that mean ozone concentration was persistently higher over Seoul (75.10 ± 7.6 ppbv) than the broader KORUS-AQ domain (70.5 ± 9.2 ppbv) at 700 hPa. Large bias reductions ($>75\%$) in the free tropospheric OH show that multiple-species assimilation is critical for balanced tropospheric chemistry analysis and emissions. The assimilation performance was dependent on the particular phase. While the evaluation of data assimilation fields shows an improved agreement with aircraft measurements in ozone (to less than 5 ppbv biases), CO, NO₂, SO₂, PAN, and OH profiles, lower tropospheric ozone analysis error was largest at stagnant conditions, whereas the model errors were mostly removed by data assimilation under dynamic weather conditions. Assimilation of new AIRS/OMI ozone profiles allowed for additional error reductions, especially under dynamic weather conditions. Our results show the important balance of dynamics and emissions both on pollution and the chemical assimilation system performance.

37. Retrievals of tropospheric ozone profiles from the synergism of AIRS and OMI: methodology and validation

Authors: Dejian Fu; Susan S. Kulawik; Kazuyuki Miyazaki; Kevin W. Bowman; John R. Worden; Annmarie Eldering; Nathaniel J. Livesey; Joao Teixeira; Fredrick W. Irion; Robert L. Herman; Gregory B. Osterman; Xiong Liu; Pieter F. Levelt; Anne M. Thompson; Ming Luo

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Abstract: . The Tropospheric Emission Spectrometer (TES) on the A-Train Aura satellite was designed to profile tropospheric ozone and its precursors, taking measurements from 2004 to 2018. Starting in 2008, TES global sampling of tropospheric ozone was gradually reduced in latitude, with global coverage stopping in 2011. To extend the record of TES, this work presents a multispectral approach that will provide O₃ data products with vertical resolution and measurement error similar to TES by combining the single-footprint thermal infrared (TIR) hyperspectral radiances from the Aqua Atmospheric Infrared Sounder (AIRS) instrument and the ultraviolet (UV) channels from the Aura Ozone Monitoring Instrument (OMI). The joint AIRS+OMI O₃ retrievals are processed through the Multi-SpEctra, Multi-SpEcies, Multi-SEnsors (MUSES) retrieval algorithm. Comparisons of collocated joint AIRS+OMI and TES to ozonesonde measurements show that both systems have similar errors, with mean and standard deviation of the differences well within the estimated measurement error. AIRS+OMI and TES have slightly different biases (within 5 parts per billion) vs. the sondes. Both AIRS and OMI have wide swath widths (~1650 km for AIRS; ~2600 km for OMI) across satellite ground tracks. Consequently, the joint AIRS+OMI measurements have the potential to maintain TES vertical sensitivity while increasing coverage by 2 orders of magnitude, thus providing an unprecedented new data set with which to quantify the evolution of tropospheric ozone.

38. Unexpected slowdown of US pollutant emission reduction in the past decade

Authors: Zhe Jiang (University of Science and Technology of China, National Center for Atmospheric Research); Brian C. McDonald (Chemical Sciences Division, National Oceanic and Atmospheric Administration Earth System Research Laboratory, Boulder, CO 80305; University of Colorado Boulder); Helen Worden (National Center for Atmospheric Research); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kazuyuki Miyazaki (Research and Development Center for Global Change, Japan Agency for Marine-Earth Science and Technology, Yokohama, 236-0001, Japan;); Zhen Qu (University of Colorado Boulder); Daven K. Henze (University of Colorado Boulder); Dylan B. A. Jones (University of Toronto); Avelino F. Arellano (University of Arizona); Emily V. Fischer (Colorado State University); Liye Zhu (Colorado State University); K. Folkert Boersma (Wageningen University, Royal Netherlands Meteorological Institute)

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Abstract: SignificanceEmissions of nitrogen oxides (NO_x) have a large impact on air quality and climate change as precursors in the formation of ozone and secondary aerosols. We find that NO_xemissions have not been decreasing as expected in recent years (2011-2015) when comparing top-down estimates from satellites and surface NO₂measurements to the trends predicted from the US Environmental Protection Agency's emission inventory data. The discrepancy can be explained by the growing relative contribution of industrial, area, and off-road mobile sources of emissions, decreasing relative contribution of on-road gasoline vehicles, and slower than expected decreases in on-road diesel NO_xemissions, with implications for air-quality management.

39. Transboundary ozone pollution across East Asia: daily evolution and photochemical production analysed by IASI+GOME2 multispectral satellite observations and models

Authors: Juan Cuesta; Yugo Kanaya; Masayuki Takigawa; Gaëlle Dufour; Maxim Eremenko; Gilles Foret; Kazuyuki Miyazaki; Matthias Beekmann

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Abstract: . We characterize a transboundary ozone pollution outbreak transported across East Asia in early May 2009 using new multispectral satellite observations of lowermost tropospheric ozone in synergy with other satellite data and models. Our analysis is focused on the daily evolution of ozone pollution plumes initially formed over the North China Plain (NCP) and their transport pathways over Northern China, Korea, Japan and the surrounding seas. A main aspect of the study is an estimation of the contribution of photochemical production of ozone along transport using the ratio of ozone to carbon monoxide enhancements with respect to background levels derived from satellite data and also from chemistry-transport models. A key contribution of the analysis is the use of new satellite data offering unprecedented skills to observe the horizontal distribution of lowermost tropospheric ozone over East Asia on daily basis, with a multispectral approach called IASI+GOME2. These satellite observations are in good agreement with ozonesondes, with low mean biases (3 %), a precision of about 16 %, a correlation coefficient of 0.85 and practically the same standard deviation for a comparison based on 2 years of data from 46 launching stations distributed worldwide, during all seasons. A similar agreement is also found over East Asia. Moreover, IASI+GOME2 offers a unique capacity for observing the evolution of near surface ozone during pollution outbreaks (with 5 % bias and 0.69 correlation), according to a comparison with surface in situ measurements during 2 major ozone events over several Japanese Islands. Single-band ozone retrievals, as those from IASI in the thermal infrared, do not capture such variability. Using IASI+GOME2, we put in evidence that (i) ozone pollution plumes are transported by an anticyclonic circulation around the Yellow Sea from the NCP to Northern China, Korea and Japan, co-located with carbon monoxide plumes, (ii) over Northern China the plume splits into two pollution filaments with one mixing with freshly emitted pollutants and (iii) ozone is produced every day of the event accounting for an enhancement in concentration during transport across East Asia of up to ~ 84 % with respect to that produced over NCP. This estimation is done according to

monotonically increasing values during 7 days of the ratio of ozone to carbon monoxide enhancements within the transported pollution plumes from about ~ 0.25 over the NCP to ~ 0.46 over the Pacific south of Japan.

40. Intercomparison of NO_x emission inventories over East Asia

Authors: Jieying Ding; Kazuyuki Miyazaki; Ronald Johannes van der A; Bas Mijling; Jun-ichi Kurokawa; SeogYeon Cho; Greet Janssens-Maenhout; Qiang Zhang; Fei Liu; Pieter Felicitas Levelt

Journal: Atmospheric Chemistry and Physics

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Abstract: . We compare nine emission inventories of nitrogen oxides including four satellite-derived NO_x inventories and the following bottom-up inventories for East Asia: REAS (Regional Emission inventory in ASia), MEIC (Multi-resolution Emission Inventory for China), CAPSS (Clean Air Policy Support System) and EDGAR (Emissions Database for Global Atmospheric Research). Two of the satellite-derived inventories are estimated by using the DECSO (Daily Emission derived Constrained by Satellite Observations) algorithm, which is based on an extended Kalman filter applied to observations from OMI or from GOME-2. The other two are derived with the EnKF algorithm, which is based on an ensemble Kalman filter applied to observations of multiple species using either the chemical transport model CHASER and MIROC-chem. The temporal behaviour and spatial distribution of the inventories are compared on a national and regional scale. A distinction is also made between urban and rural areas. The intercomparison of all inventories shows good agreement in total NO_x emissions over mainland China, especially for trends, with an average bias of about 20 % for yearly emissions. All the inventories show the typical emission reduction of 10 % during the Chinese New Year and a peak in December. Satellite-derived approaches using OMI show a summer peak due to strong emissions from soil and biomass burning in this season. Biases in NO_x emissions and uncertainties in temporal variability increase quickly when the spatial scale decreases. The analyses of the differences show the importance of using observations from multiple instruments and a high spatial resolution model for the satellite-derived inventories, while for bottom-up inventories, accurate emission factors and activity information are required. The advantage of the satellite-derived approach is that the emissions are soon available after observation, while the strength of the bottom-up inventories is that they include detailed information of emissions for each source category.

41. Seasonal and spatial changes in trace gases over megacities from Aura TES observations: two case studies

Authors: Karen E. Cady-Pereira; Vivienne H. Payne; Jessica L. Neu; Kevin W. Bowman; Kazuyuki Miyazaki; Eloise A. Marais; Susan Kulawik; Zitely A. Tzompa-Sosa; Jennifer D. Hegarty

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Abstract: . The Aura Tropospheric Emission Spectrometer (TES) is collecting closely spaced observations over 19 megacities. The objective is to obtain measurements that will lead to better understanding of the processes affecting air quality in and around these cities, and to better estimates of the seasonal and interannual variability. We explore the TES measurements of ozone, ammonia, methanol and formic acid collected around the Mexico City metropolitan area (MCMA) and in the vicinity of Lagos (Nigeria). The TES data exhibit seasonal signals that are correlated with Atmospheric Infrared Sounder (AIRS) CO and Moderate Resolution Imaging Spectroradiometer (MODIS) aerosol optical depth (AOD), with in situ measurements in the MCMA and with Goddard Earth Observing System (GEOS)-Chem model output in the Lagos area. TES was able to detect an extreme pollution event in the MCMA on 9 April 2013, which is also evident in the in situ data. TES data also show that biomass burning has a greater impact south of the city than in the caldera where Mexico City is located. TES measured enhanced values of the four species over the Gulf of Guinea south of Lagos. Since it observes many cities from the same platform with the same instrument and applies the same retrieval algorithms, TES data provide a very useful tool for easily comparing air quality measures of two or more cities. We compare the data from the MCMA and Lagos, and show that, while the MCMA has occasional extreme pollution events, Lagos consistently has higher levels of these trace gases.

42. Evaluation of ACCMIP ozone simulations and ozonesonde sampling biases using a satellite-based multi-constituent chemical reanalysis

Authors: Kazuyuki Miyazaki; Kevin Bowman

Journal: Atmospheric Chemistry and Physics

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Abstract: . The Atmospheric Chemistry Climate Model Intercomparison Project (ACCMIP) ensemble ozone simulations for the present day from the 2000 decade simulation results are evaluated by a state-of-the-art multi-constituent atmospheric chemical reanalysis that ingests multiple satellite data including the Tropospheric Emission Spectrometer (TES), the Microwave Limb Sounder (MLS), the Ozone Monitoring Instrument (OMI), and the Measurement of Pollution in the Troposphere (MOPITT) for 2005-2009. Validation of the chemical reanalysis against global ozonesondes shows good agreement throughout the free troposphere and lower stratosphere for both seasonal and year-to-year variations, with an annual mean bias of less than 0.9 ppb in the middle and upper troposphere at the tropics and mid-latitudes. The reanalysis provides comprehensive spatiotemporal evaluation of chemistry-model performance that compliments direct ozonesonde comparisons, which are shown to suffer from significant sampling bias. The reanalysis reveals that the ACCMIP ensemble mean overestimates ozone in the northern extratropics by 6-11 ppb while underestimating by up to 18 ppb in the southern tropics over the Atlantic in the lower troposphere. Most models underestimate the spatial variability of the annual mean lower tropospheric concentrations in the extratropics of both hemispheres by up to 70 %. The ensemble mean also overestimates the seasonal amplitude by 25-70 % in the northern extratropics and overestimates the inter-hemispheric gradient by about 30 % in the lower and middle troposphere. A part of the discrepancies can be attributed to the 5-year reanalysis data for the decadal model simulations. However, these differences are less evident with the current sonde network. To estimate ozonesonde sampling biases, we computed model bias separately for global coverage and the ozonesonde network. The ozonesonde sampling bias in the evaluated model bias for the seasonal mean concentration relative to global coverage is 40-50 % over the western Pacific and east Indian Ocean and reaches 110 % over the equatorial Americas and up to 80 % for the global tropics. In contrast, the ozonesonde sampling bias is typically smaller than 30 % for the Arctic regions in the lower and middle troposphere. These systematic biases have implications for ozone radiative forcing and the response of chemistry to climate that can be further quantified as the satellite observational record extends to multiple decades.

43. Decadal changes in global surface NO_x emissions from multi-constituent satellite data assimilation

Authors: Kazuyuki Miyazaki; Henk Eskes; Kengo Sudo; K. Folkert Boersma; Kevin Bowman; Yugo Kanaya

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Abstract: . Global surface emissions of nitrogen oxides (NO_x) over a 10-year period (2005-2014) are estimated from an assimilation of multiple satellite data sets: tropospheric NO₂ columns from Ozone Monitoring Instrument (OMI), Global Ozone Monitoring Experiment-2 (GOME-2), and Scanning Imaging Absorption Spectrometer for Atmospheric Chartography (SCIAMACHY), O₃ profiles from Tropospheric Emission Spectrometer (TES), CO profiles from Measurement of Pollution in the Troposphere (MOPITT), and O₃ and HNO₃ profiles from Microwave Limb Sounder (MLS) using an ensemble Kalman filter technique. Chemical concentrations of various species and emission sources of several precursors are simultaneously optimized. This is expected to improve the emission inversion because the emission estimates are influenced by biases in the modelled tropospheric chemistry, which can be partly corrected by also optimizing the concentrations. We present detailed distributions of the estimated emission distributions for all major regions, the diurnal and seasonal variability, and the evolution of these emissions over the 10-year period. The estimated regional total emissions show a strong positive trend over India (+29 % decade⁻¹), China (+26 % decade⁻¹), and the Middle East (+20 % decade⁻¹), and a negative trend over the USA (-38 % decade⁻¹), southern Africa (-8.2 % decade⁻¹), and western Europe (-8.8 % decade⁻¹). The negative trends in the USA and western Europe are larger during 2005-2010 relative to 2011-2014, whereas the trend in China becomes negative after 2011. The data assimilation also suggests a large uncertainty in anthropogenic and fire-related emission factors and an important underestimation of soil NO_x sources in the emission inventories. Despite the large trends observed for individual regions, the global total emission is almost constant between 2005 (47.9 Tg N yr⁻¹) and 2014 (47.5 Tg N yr⁻¹).

44. Impacts of anthropogenic and natural sources on free tropospheric ozone over the Middle East

Authors: Zhe Jiang; Kazuyuki Miyazaki; John R. Worden; Jane J. Liu; Dylan B. A. Jones; Daven K. Henze

Journal: Atmospheric Chemistry and Physics

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Abstract: . Significant progress has been made in identifying the influence of different processes and emissions on the summertime enhancements of free tropospheric ozone (O₃) at northern midlatitude regions. However, the exact contribution of regional emissions, chemical and transport processes to these summertime enhancements is still not well quantified. Here we focus on quantifying the influence of regional emissions on the summertime O₃ enhancements over the Middle East, using updated reactive nitrogen (NO_x) emissions. We then use the adjoint of the GEOS-Chem model with these updated NO_x emissions to show that the global total contribution of lightning NO_x on middle free tropospheric O₃ over the Middle East is about 2 times larger than that from global anthropogenic sources. The summertime middle free tropospheric O₃ enhancement is primarily due to Asian NO_x emissions, with approximately equivalent contributions from Asian anthropogenic activities and lightning. In the Middle Eastern lower free troposphere, lightning NO_x from Europe and North America and anthropogenic NO_x from Middle Eastern local emissions are the primary sources of O₃. This work highlights the critical role of lightning NO_x on northern midlatitude free tropospheric O₃ and the important effect of the Asian summer monsoon on the export of Asian pollutants.

45. A tropospheric chemistry reanalysis for the years 2005-2012 based on an assimilation of OMI, MLS, TES, and MOPITT satellite data

Authors: K. Miyazaki; H. J. Eskes; K. Sudo

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Abstract: . We present the results from an 8-year tropospheric chemistry reanalysis for the period 2005-2012 obtained by assimilating multiple data sets from the OMI, MLS, TES, and MOPITT satellite instruments. The reanalysis calculation was conducted using a global chemical transport model and an ensemble Kalman filter technique that simultaneously optimises the chemical concentrations of various species and emissions of several precursors. The optimisation of both the concentration and the emission fields is an efficient method to correct the entire tropospheric profile and its year-to-year variations, and to adjust various tracers chemically linked to the species assimilated. Comparisons against independent aircraft, satellite, and ozonesonde observations demonstrate the quality of the analysed O₃, NO₂, and CO concentrations on regional and global scales and for both seasonal and year-to-year variations from the lower troposphere to the lower stratosphere. The data assimilation statistics imply persistent reduction of model error and improved representation of emission variability, but they also show that discontinuities in the availability of the measurements lead to a degradation of the reanalysis. The decrease in the number of assimilated measurements increased the ozonesonde-minus-analysis difference after 2010 and caused spurious variations in the estimated emissions. The Northern/Southern Hemisphere OH ratio was modified considerably due to the multiple-species assimilation and became closer to an observational estimate, which played an important role in propagating observational information among various chemical fields and affected the emission estimates. The consistent concentration and emission products provide unique information on year-to-year variations in the atmospheric environment.